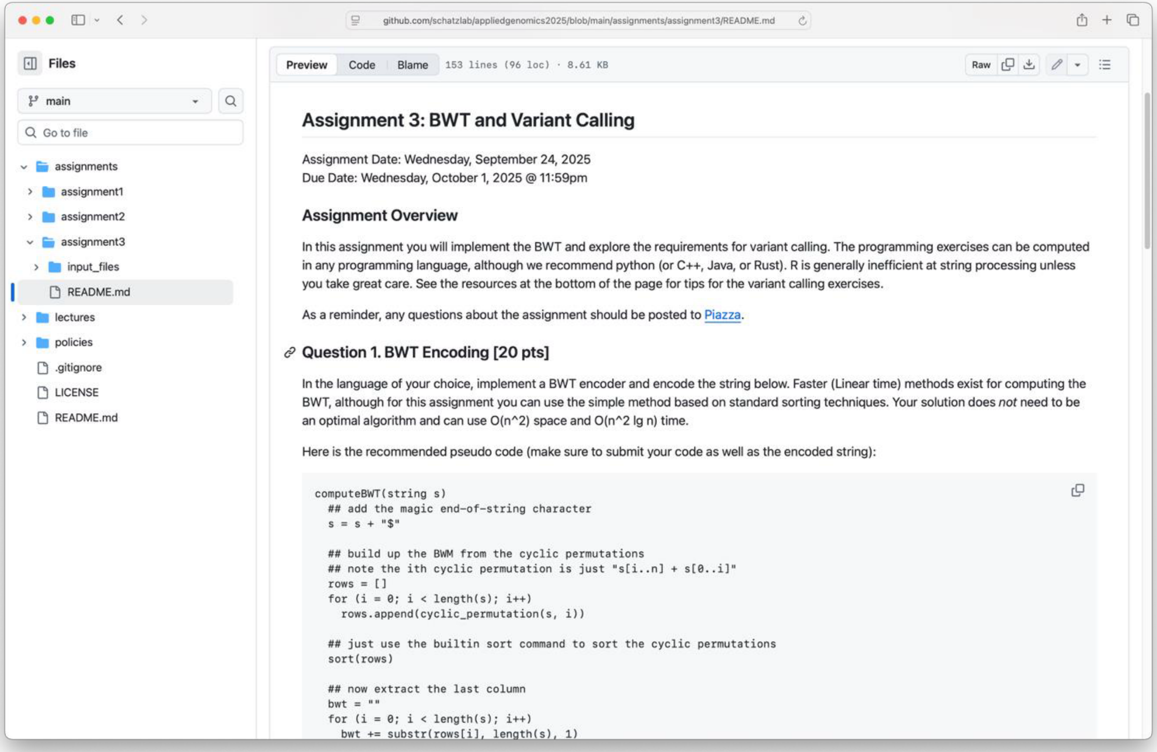
Introduction to Machine Learning

Michael Schatz
September 29, 2025

Assignment 3: Variant Calling
Due Wed Oct 1 by 11:59pm



github.com/schatzlab/appliedgenomics2025/blob/main/assignments/assignment3/README.md

Files

main

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Preview | Code | Blame | 153 lines (96 loc) · 8.61 KB | Raw

Assignment 3: BWT and Variant Calling

Assignment Date: Wednesday, September 24, 2025
Due Date: Wednesday, October 1, 2025 @ 11:59pm

Assignment Overview

In this assignment you will implement the BWT and explore the requirements for variant calling. The programming exercises can be computed in any programming language, although we recommend python (or C++, Java, or Rust). R is generally inefficient at string processing unless you take great care. See the resources at the bottom of the page for tips for the variant calling exercises.

As a reminder, any questions about the assignment should be posted to Piazza.

Question 1. BWT Encoding [20 pts]

In the language of your choice, implement a BWT encoder and encode the string below. Faster (Linear time) methods exist for computing the BWT, although for this assignment you can use the simple method based on standard sorting techniques. Your solution does *not* need to be an optimal algorithm and can use O(n^2) space and O(n^2 lg n) time.

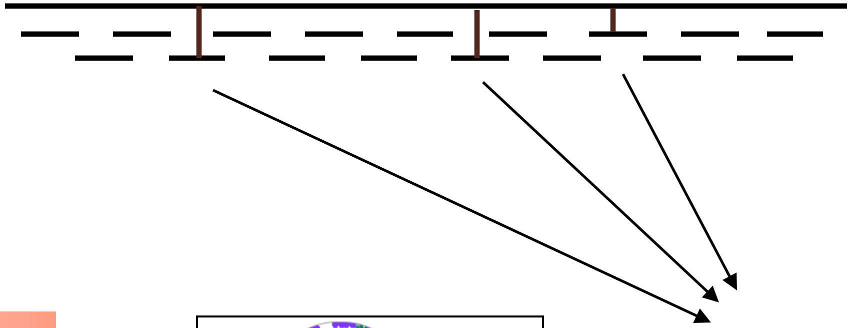
Here is the recommended pseudo code (make sure to submit your code as well as the encoded string):

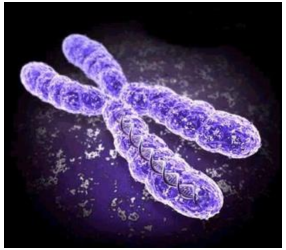
```
computeBWT(string s)
  ## add the magic end-of-string character
  s = s + "$"

  ## build up the BWM from the cyclic permutations
  ## note the ith cyclic permutation is just "s[i..n] + s[0..i]"
  rows = []
  for (i = 0; i < length(s); i++)
    rows.append(cyclic_permutation(s, i))

  ## just use the builtin sort command to sort the cyclic permutations
  sort(rows)

  ## now extract the last column
  bwt = ""
  for (i = 0; i < length(s); i++)
    bwt += substr(rows[i], length(s), 1)
```

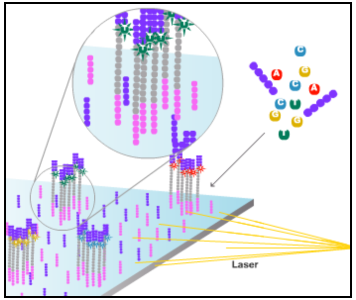
https://github.com/schatzlab/appliedgenomics2025/tree/main/assignments/assignment3
Check Piazza for questions!

Personal Genomics

How does your genome compare to the reference?







Heart Disease

Cancer

Amazing actress

Exact Matching Review & Overview

Where is GATTACA in the human genome?

Brute Force
(3 GB)

```
BANANA
BAN
 ANA
  NAN
   ANA
```

O(m * n)

Slow & Easy

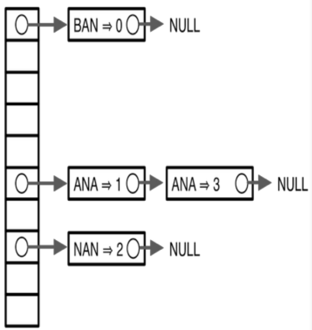
Suffix Array
(>15 GB)

6	$
5	A$
3	ANA$
1	ANANA$
0	BANANA$
4	NA$
2	NANA$

O(m + lg n)

Full-text index

Hash Table
(>15 GB)



O(1)

Fixed-length lookup

BWT
(3 GB)

BANANA$
=>
$BANAN**A**
A$BANA**N**
ANA$BA**N**
ANANA$**B**
BANANA**$**
NA$BAN**A**
NANA$B**A**
=>
ANNB$AA

O(m)

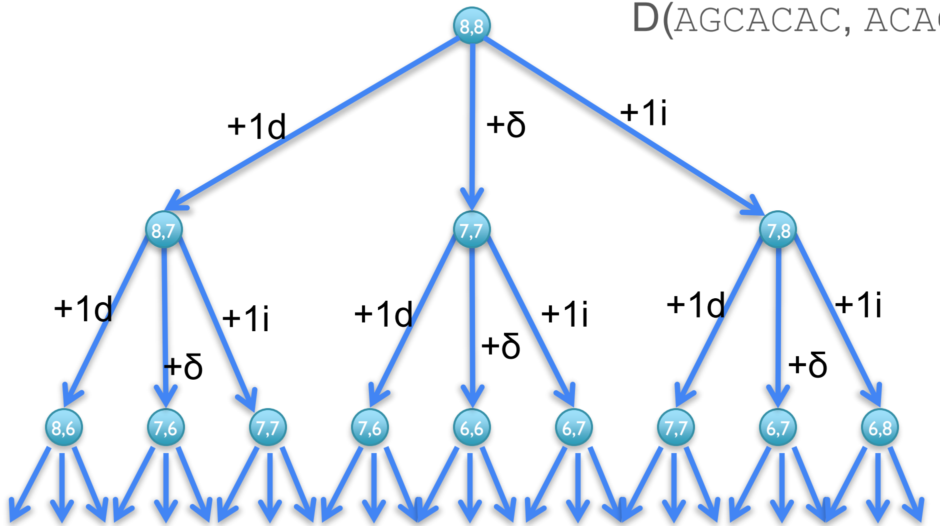
Full-text and concise

*** These are general techniques applicable to any text search problem ***

Recursive solution

- Computation of D is a recursive process.
  - At each step, we only allow matches, substitutions, and indels
  - D(i,j) in terms of D(i',j') for i' ≤ i and j' ≤ j.

D(AGCACACA, ACACACTA) = min{D(AGCACACA, ACACACT) + 1,
D(AGCACAC, ACACACTA) + 1,
D(AGCACAC, ACACACT) +δ(A, A)}



[What is the running time?]

Dynamic Programming Matrix

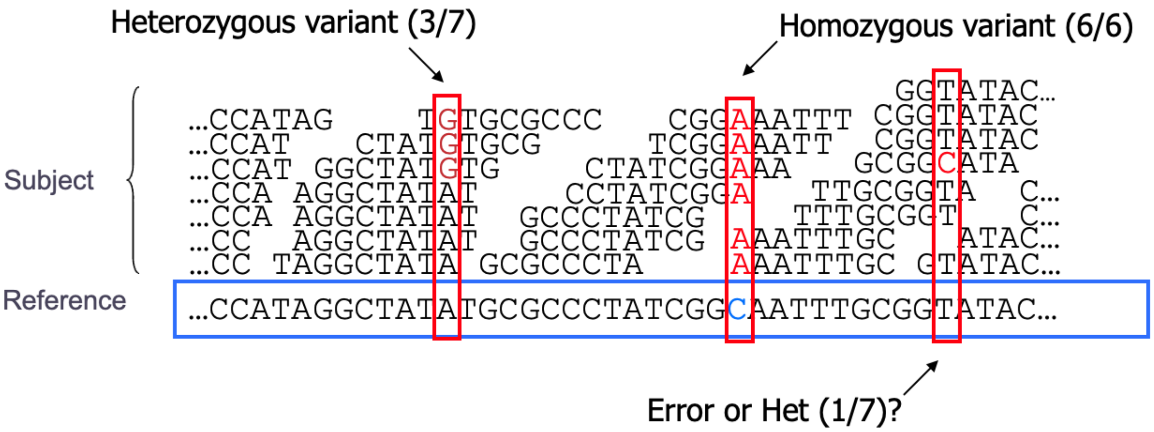
		A	C	A	C	A	C	T	A
	0	1	2	3	4	5	6	7	8
A	1	0	1	2	3	4	5	6	7
G	2	1	1	2	3	4	5	6	7
C	3	2	1	2	2	3	4	5	6
A	4	3	2	1	2	2	3	4	5
C	5	4	3	2	1	2	2	3	4
A	6	5	4	3	2	1	2	3	3
C	7	6	5	4	3	2	1	2	3
A	8	7	6	5	4	3	2	2	2

D[AGCACACA,ACACACTA] = 2

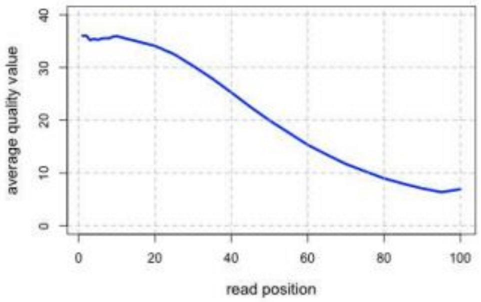
```
AGCACAC-A
|*|||||*|
A-CACACTA
```

[Can we do it any better?]

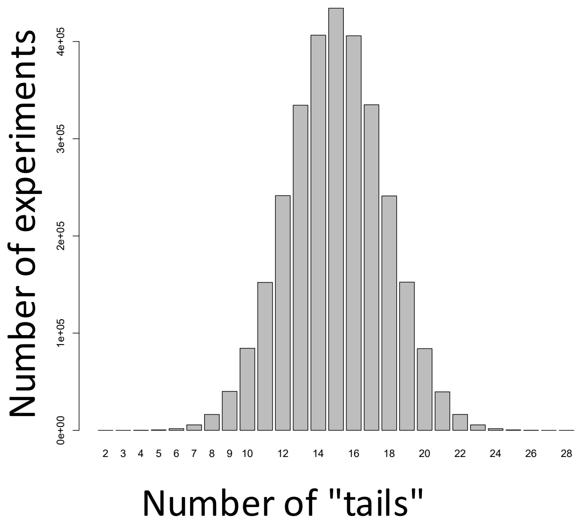
Genotyping Theory



- If there were no sequencing errors, identifying SNPs would be very easy: any time a read disagrees with the reference, it must be a variant!
- Sequencing instruments make mistakes
  - Quality of read decreases over the read length
- A single read differing from the reference is probably just an error, but it becomes more likely to be real as we see it multiple times



So, with 30 tosses (reads), we are much more likely to see an even mix of alternate and reference alleles at a heterozygous locus in a genome



This is why <u>at least</u> a "30X" (30 fold sequence coverage) genome is recommended: it confers sufficient power to distinguish heterozygous alleles and from mere sequencing errors

P(3/30 het) <?> P(3/30 err)

PolyBayes: The first statistically rigorous variant detection tool.

letter © 1999 Nature America Inc. • http://genetics.nature.com

A general approach to single-nucleotide polymorphism discovery

Gabor T. Marth[1], Ian Korf[1], Mark D. Yandell[1], Raymond T. Yeh[1], Zhijie Gu[2], Hamideh Zakeri[2], Nathan O. Stitziel[1], LaDeana Hillier[1], Pui-Yan Kwok[2] & Warren R. Gish[1]

Bayesian posterior probability

Base call + Base quality

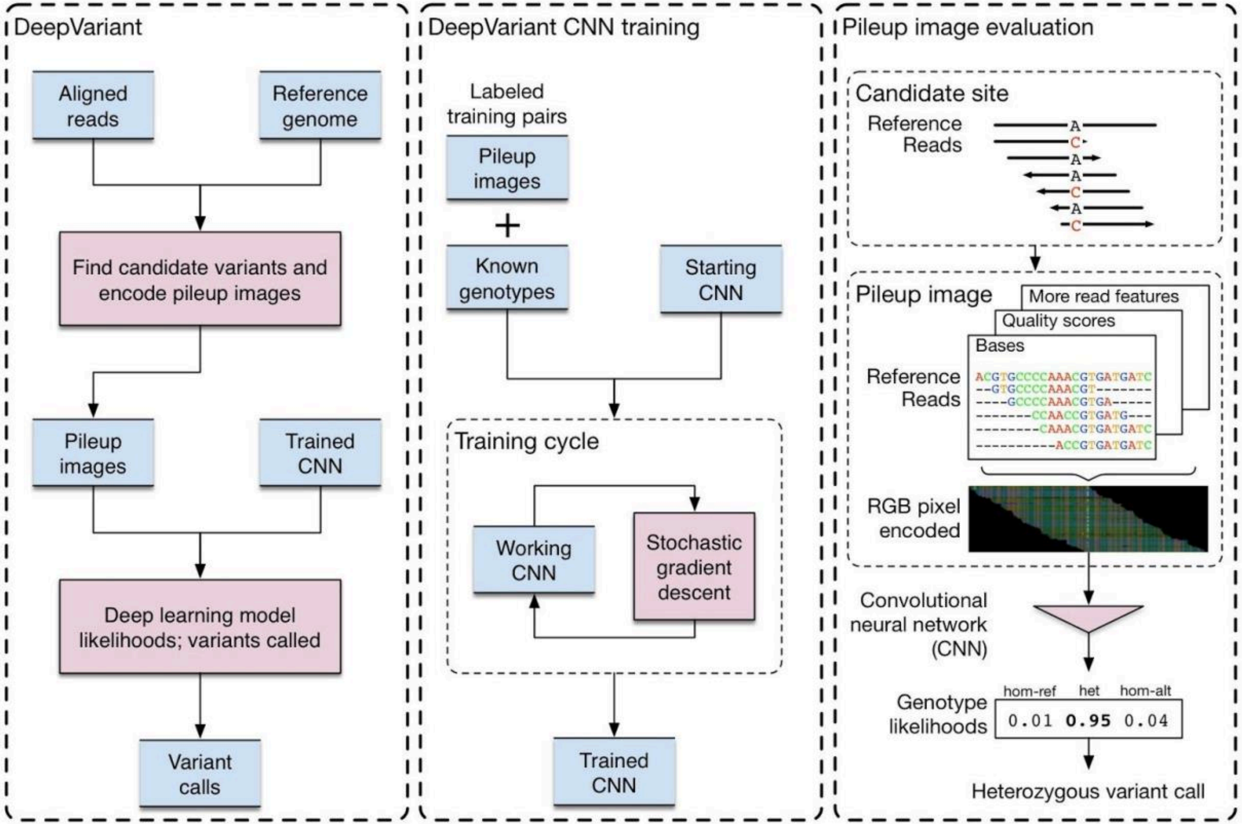
Expected (prior) polymorphism rate

$$\boxed{P(SNP)=} \sum_{all\ variable\ S} \frac{\boxed{\frac{P(S_1|R_1)}{P_{Prior}(S_1)} \cdot \ldots \cdot \frac{P(S_N|R_N)}{P_{Prior}(S_N)}} \cdot \boxed{P_{Prior}(S_1,\ldots,S_N)}}{\sum_{S_{i_1}\in[A,C,G,T]} \cdots \sum_{S_{i_N}\in[A,C,G,T]} \boxed{\frac{P(S_{i_1}|R_1)}{P_{Prior}(S_{i_1})}} \cdot \ldots \cdot \frac{P(S_{i_N}|R_1)}{P_{Prior}(S_{i_N})} \cdot P_{Prior}(S_{i_1},\ldots,S_{i_N})}$$

Probability of observed base composition (should model sequencing error rate)

http://www.nature.com/ng/journal/v23/n4/full/ng1299_452.html

DeepVariant

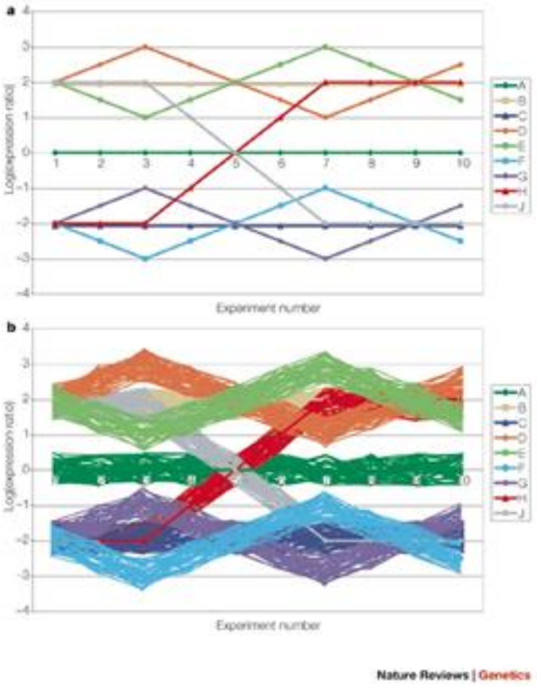


Creating a universal SNP and small indel variant caller with deep neural networks

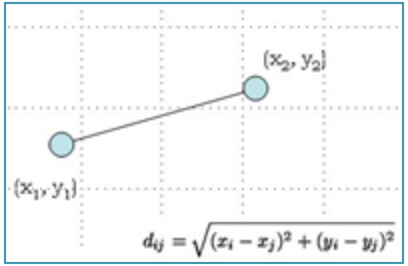
Poplin et al. (2018) Nature Biotechnology. https://www.nature.com/articles/nbt.4235

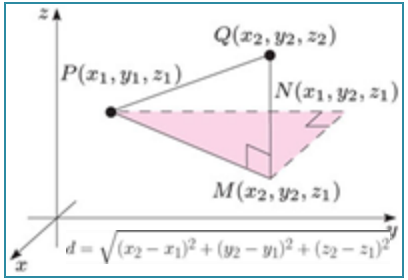
Clustering & Dimensionality Reduction

Clustering Refresher



Euclidean Distance



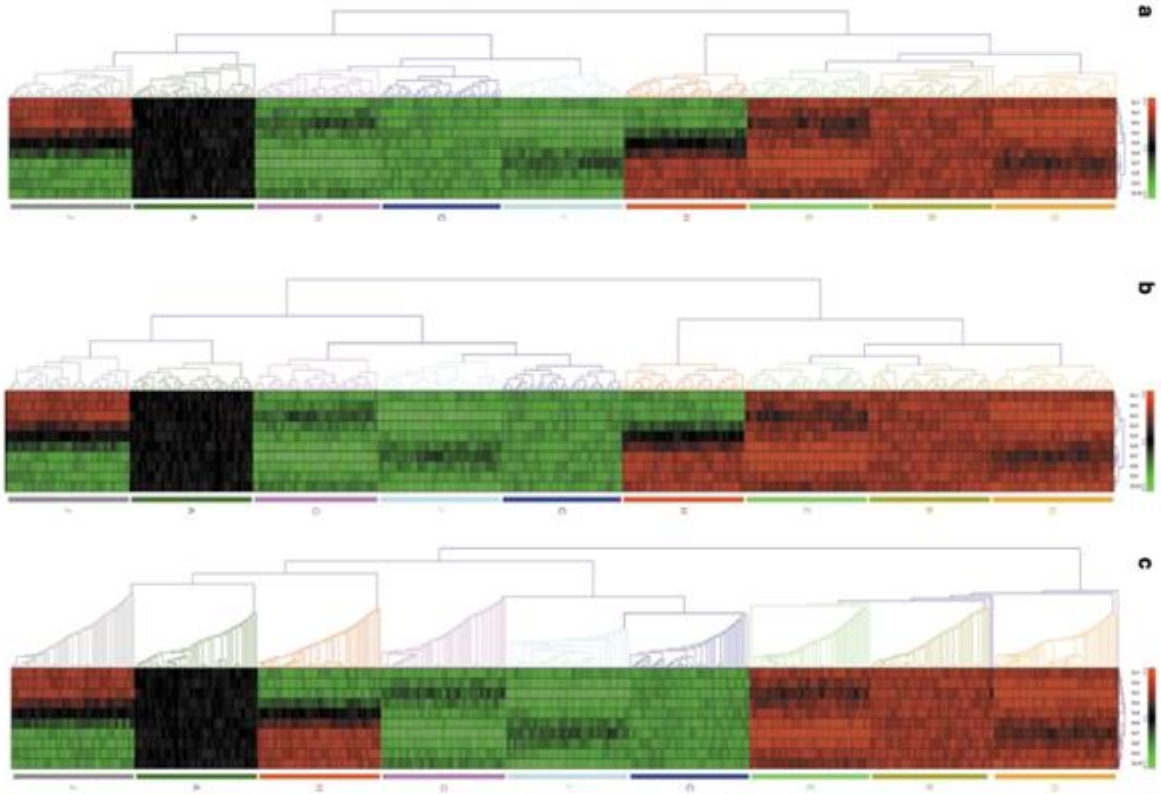


$$\mathrm{d}(\mathbf{p}, \mathbf{q}) = \mathrm{d}(\mathbf{q}, \mathbf{p}) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2 + \cdots + (q_n - p_n)^2} = \sqrt{\sum_{i=1}^{n} (q_i - p_i)^2}.$$

Computational genetics: Computational analysis of microarray data

Quackenbush (2001) *Nature Reviews Genetics.* doi:10.1038/35076576

Hierarchical Clustering

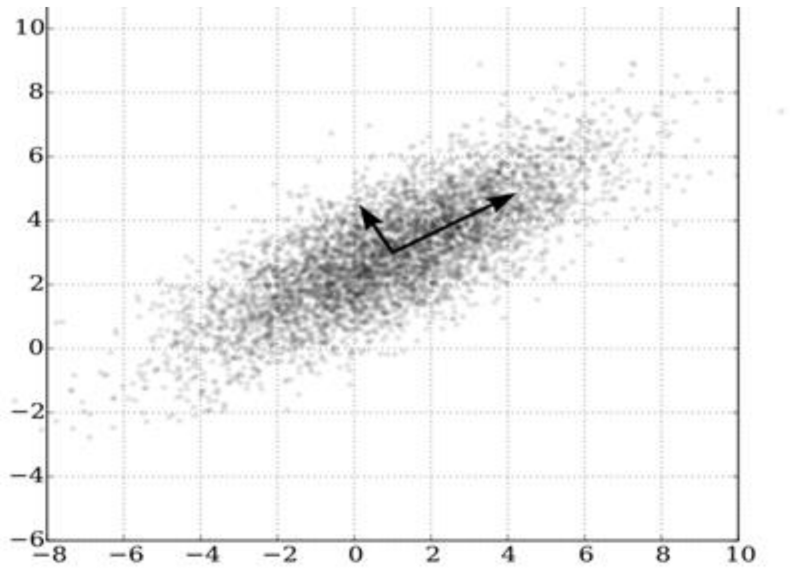


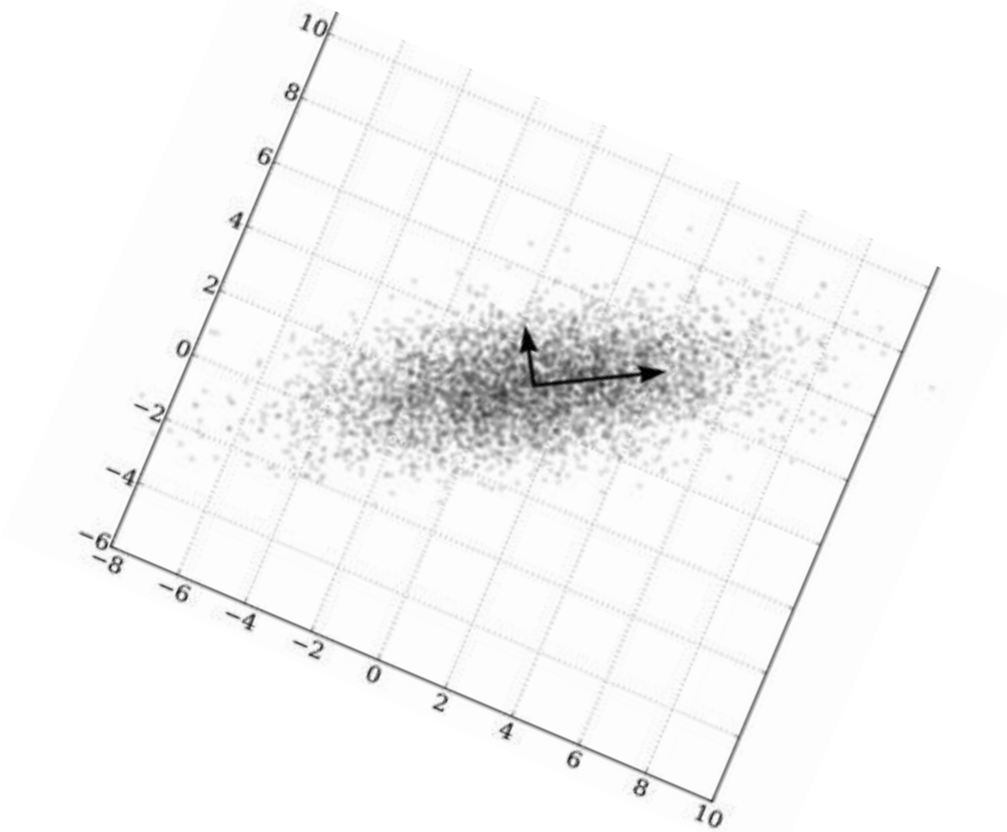
average

complete

single

Principal Components Analysis (PCA)





PC1: “New X”- The dimension with the most variability

PC2: “New Y”- The dimension with the second most variability

Principal Components Analysis (PCA)

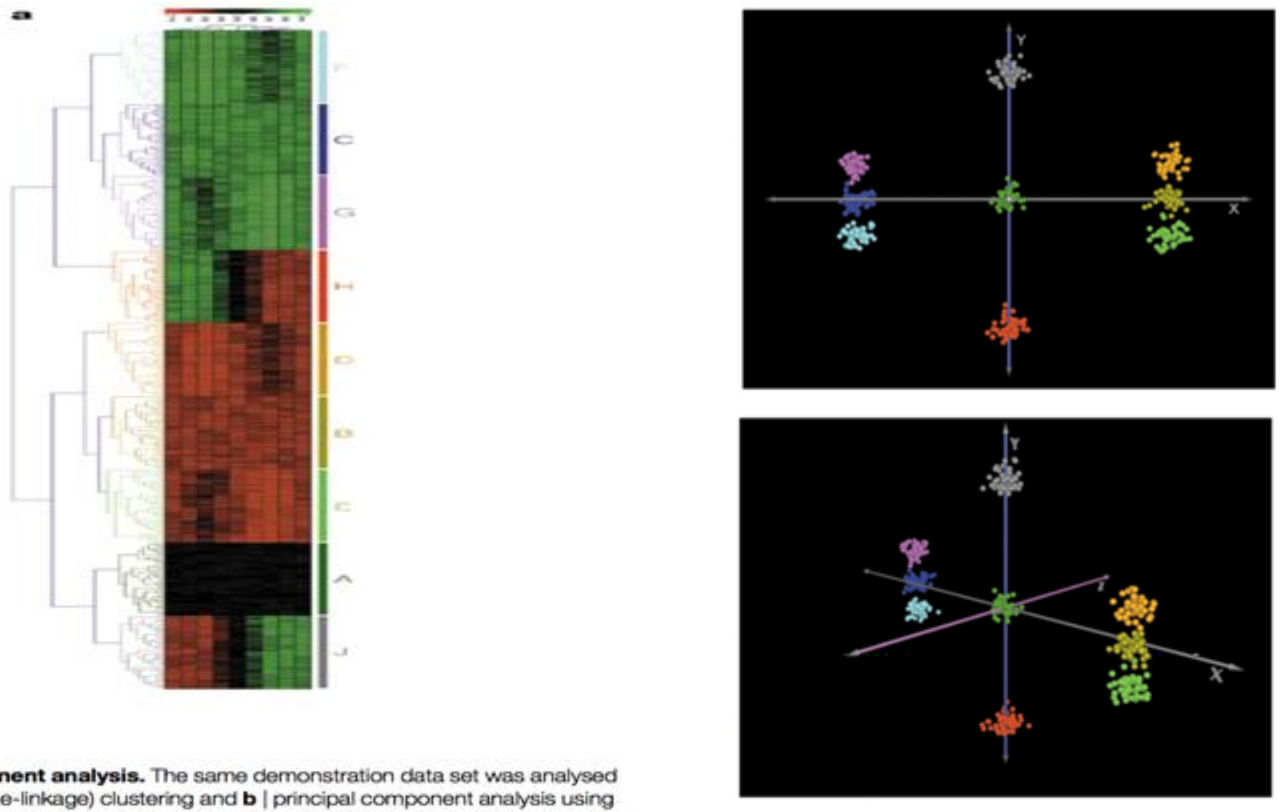


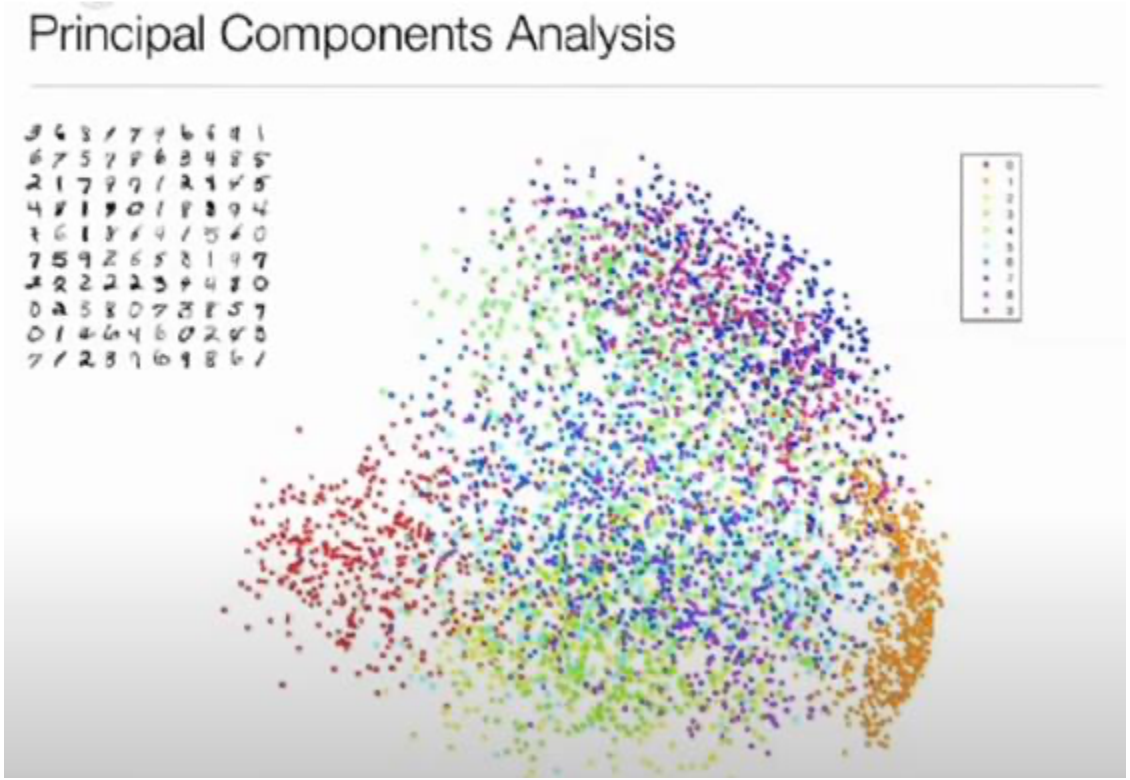
Figure 4 | **Principal component analysis.** The same demonstration data set was analysed using **a** | hierarchical (average-linkage) clustering and **b** | principal component analysis using Euclidean distance, to show how each treats the data, with genes colour coded on the basis of hierarchical clustering results for comparison.

MNIST

- 70,000 images, each labeled with the correct digit (0-9)
    - 28 x 28 thumbnail images of handwritten digits
    - Grayscale image with 256 possible values (784 bytes per image)

- Widely used dataset for image classification
    - Very simple, the categories are very easy to understand

MNIST: PCA



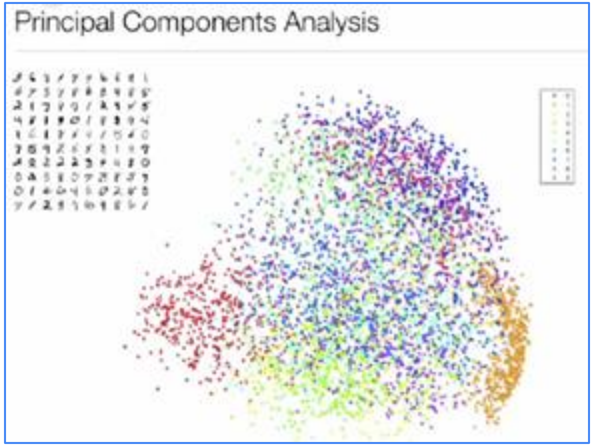
Each 28x28 image can be considered a point in 784 dimensional space!

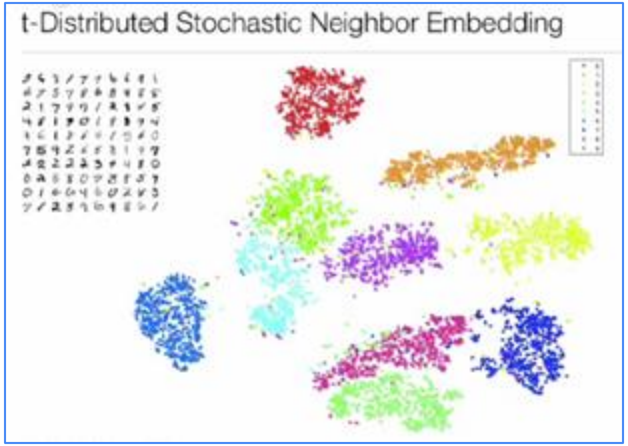
PCA roughly groups digits together

- 0 (red) in bottom left
- 1 (orange) in bottom right

But very incomplete separation, lots of mixing of colors

MNIST: PCA and t-SNE





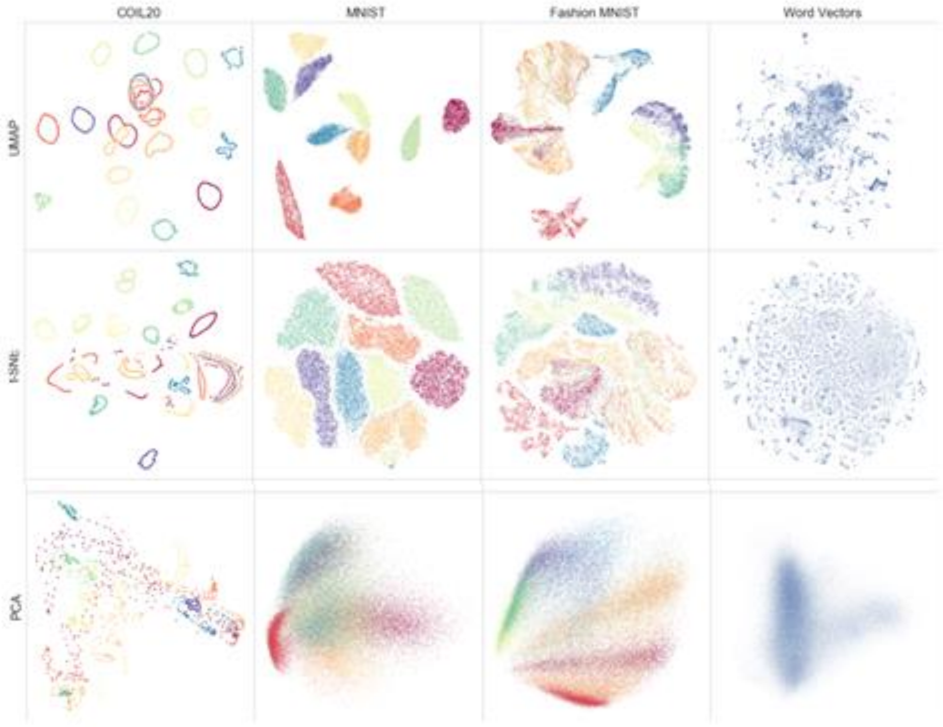
t-distributed Stochastic Neighborhood Embedding

- Non-linear dimensionality reduction technique: distances are only locally meaningful
- Rather than Euclidean distances, for each point fits a Gaussian kernel to fit the nearest N neighbors (perplexity) that define the probabilities that two points should be close together
- Using an iterative spring embedding system to place high probability points nearby

Visualizing Data Using t-SNE

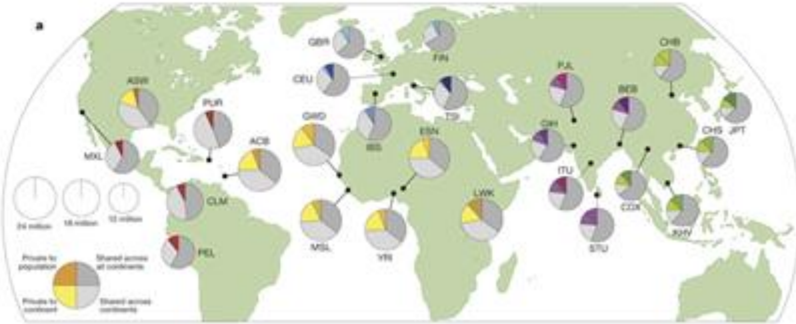
https://www.youtube.com/watch?v=RJVL80Gg3IA

UMAP

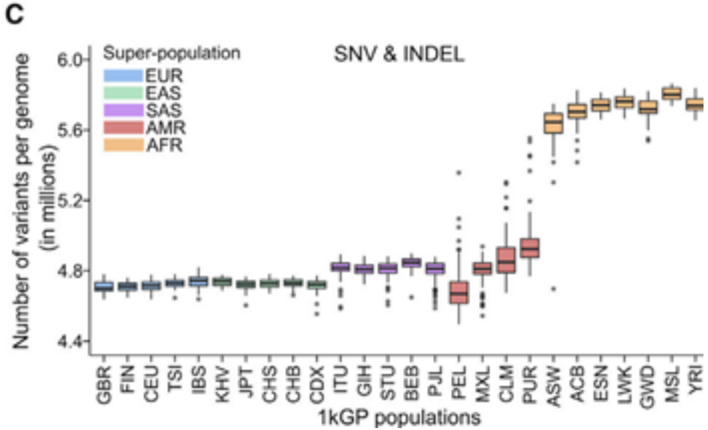


UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction
McInnes et al (2018) arXiv. 1802.03426
https://www.youtube.com/watch?v=nq6iPZVUxZU
https://towardsdatascience.com/how-exactly-umap-works-13e3040e1668

1000 Genomes Project

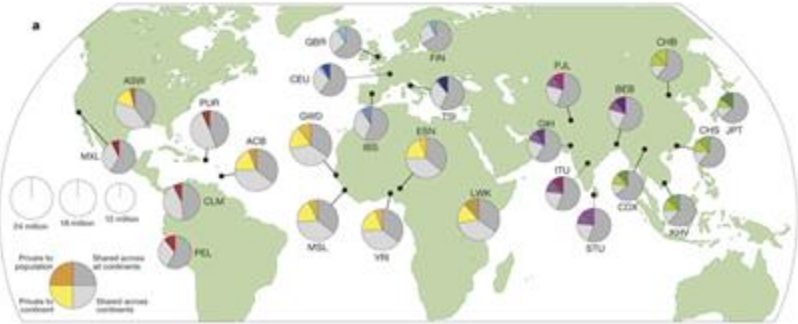


EAS	CHB	Han Chinese	Han Chinese in Beijing, China
	JPT	Japanese	Japanese in Tokyo, Japan
	CHS	Southern Han Chinese	Han Chinese South
	CDX	Dai Chinese	Chinese Dai in Xishuangbanna, China
	KHV	Kinh Vietnamese	Kinh in Ho Chi Minh City, Vietnam
	CHD	Denver Chinese	Chinese in Denver, Colorado (pilot 3 only)
EUR	CEU	CEPH	Utah residents (CEPH) with Northern and Western European ancestry
	TSI	Tuscan	Toscani in Italia
	GBR	British	British in England and Scotland
	FIN	Finnish	Finnish in Finland
	IBS	Spanish	Iberian populations in Spain
AFR	YRI	Yoruba	Yoruba in Ibadan, Nigeria
	LWK	Luhya	Luhya in Webuye, Kenya
	GWD	Gambian	Gambian in Western Division, The Gambia
	MSL	Mende	Mende in Sierra Leone
	ESN	Esan	Esan in Nigeria
AMR	ASW	African-American SW	African Ancestry in Southwest US
	ACB	African-Caribbean	African Caribbean in Barbados
	MXL	Mexican-American	Mexican Ancestry in Los Angeles, California
	PUR	Puerto Rican	Puerto Rican in Puerto Rico
	CLM	Colombian	Colombian in Medellin, Colombia
	PEL	Peruvian	Peruvian in Lima, Peru
SAS	GIH	Gujarati	Gujarati Indian in Houston, TX
	PJL	Punjabi	Punjabi in Lahore, Pakistan
	BEB	Bengali	Bengali in Bangladesh
	STU	Sri Lankan	Sri Lankan Tamil in the UK
	ITU	Indian	Indian Telugu in the UK



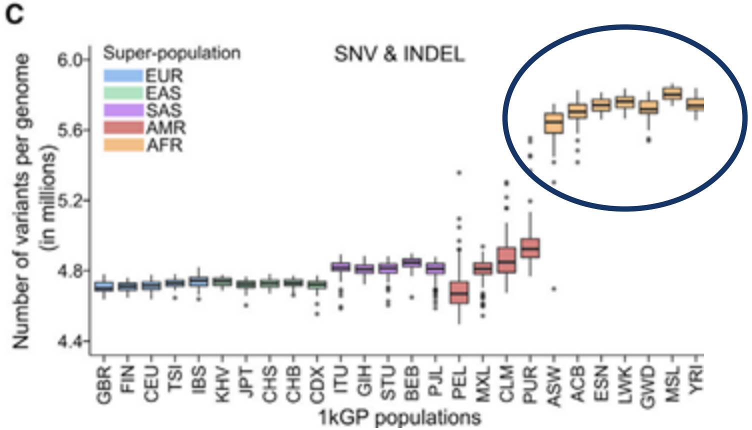
High-coverage whole-genome sequencing of the expanded 1000 Genomes Project cohort including 602 trios
Byrska-Bishop et al. (2022) Cell. doi: 10.1016/j.cell.2022.08.004

1000 Genomes Project



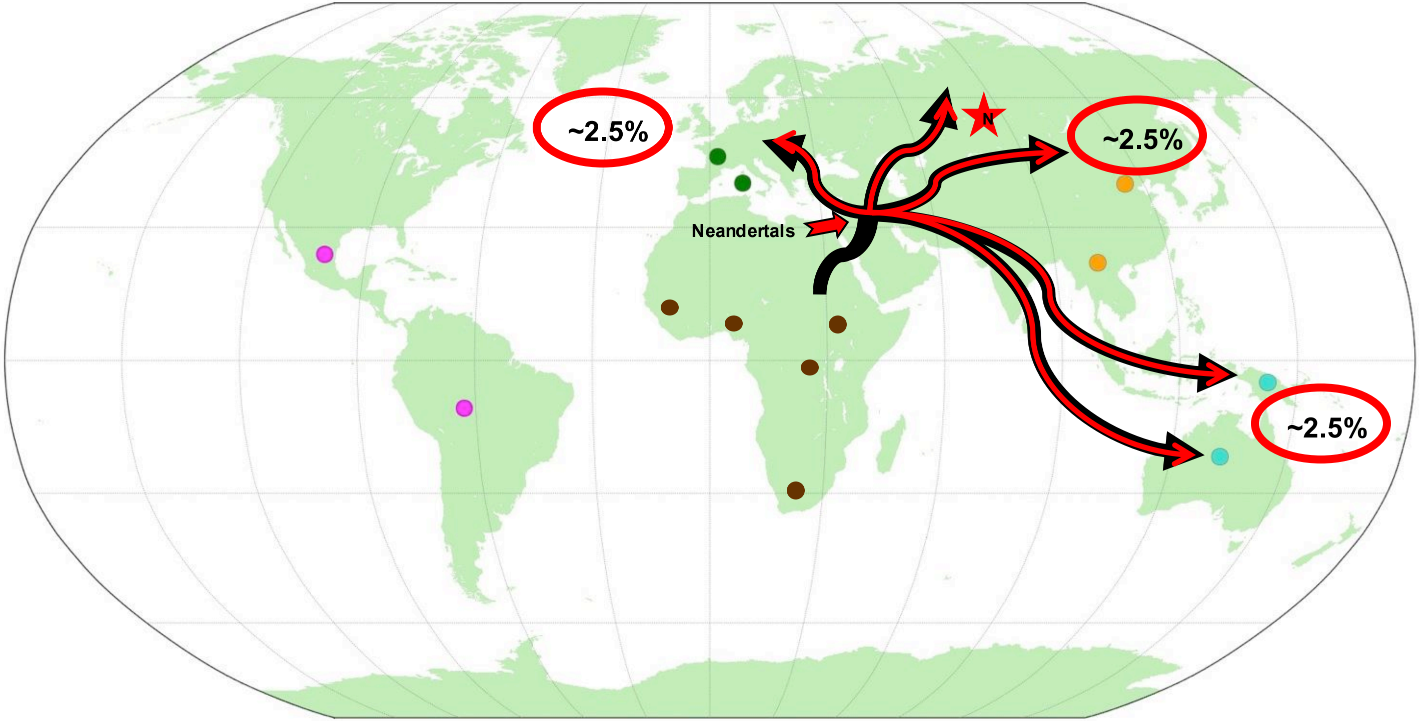
Why do people with African ancestry have so many more variants?

EAS	CHB	Han Chinese	Han Chinese in Beijing, China
	JPT	Japanese	Japanese in Tokyo, Japan
	CHS	Southern Han Chinese	Han Chinese South
	CDX	Dai Chinese	Chinese Dai in Xishuangbanna, China
	KHV	Kinh Vietnamese	Kinh in Ho Chi Minh City, Vietnam
	CHD	Denver Chinese	Chinese in Denver, Colorado (pilot 3 only)
EUR	CEU	CEPH	Utah residents (CEPH) with Northern and Western European ancestry
	TSI	Tuscan	Toscani in Italia
	GBR	British	British in England and Scotland
	FIN	Finnish	Finnish in Finland
	IBS	Spanish	Iberian populations in Spain
AFR	YRI	Yoruba	Yoruba in Ibadan, Nigeria
	LWK	Luhya	Luhya in Webuye, Kenya
	GWD	Gambian	Gambian in Western Division, The Gambia
	MSL	Mende	Mende in Sierra Leone
	ESN	Esan	Esan in Nigeria
AMR	ASW	African-American SW	African Ancestry in Southwest US
	ACB	African-Caribbean	African Caribbean in Barbados
	MXL	Mexican-American	Mexican Ancestry in Los Angeles, California
	PUR	Puerto Rican	Puerto Rican in Puerto Rico
	CLM	Colombian	Colombian in Medellin, Colombia
	PEL	Peruvian	Peruvian in Lima, Peru
SAS	GIH	Gujarati	Gujarati Indian in Houston, TX
	PJL	Punjabi	Punjabi in Lahore, Pakistan
	BEB	Bengali	Bengali in Bangladesh
	STU	Sri Lankan	Sri Lankan Tamil in the UK
	ITU	Indian	Indian Telugu in the UK

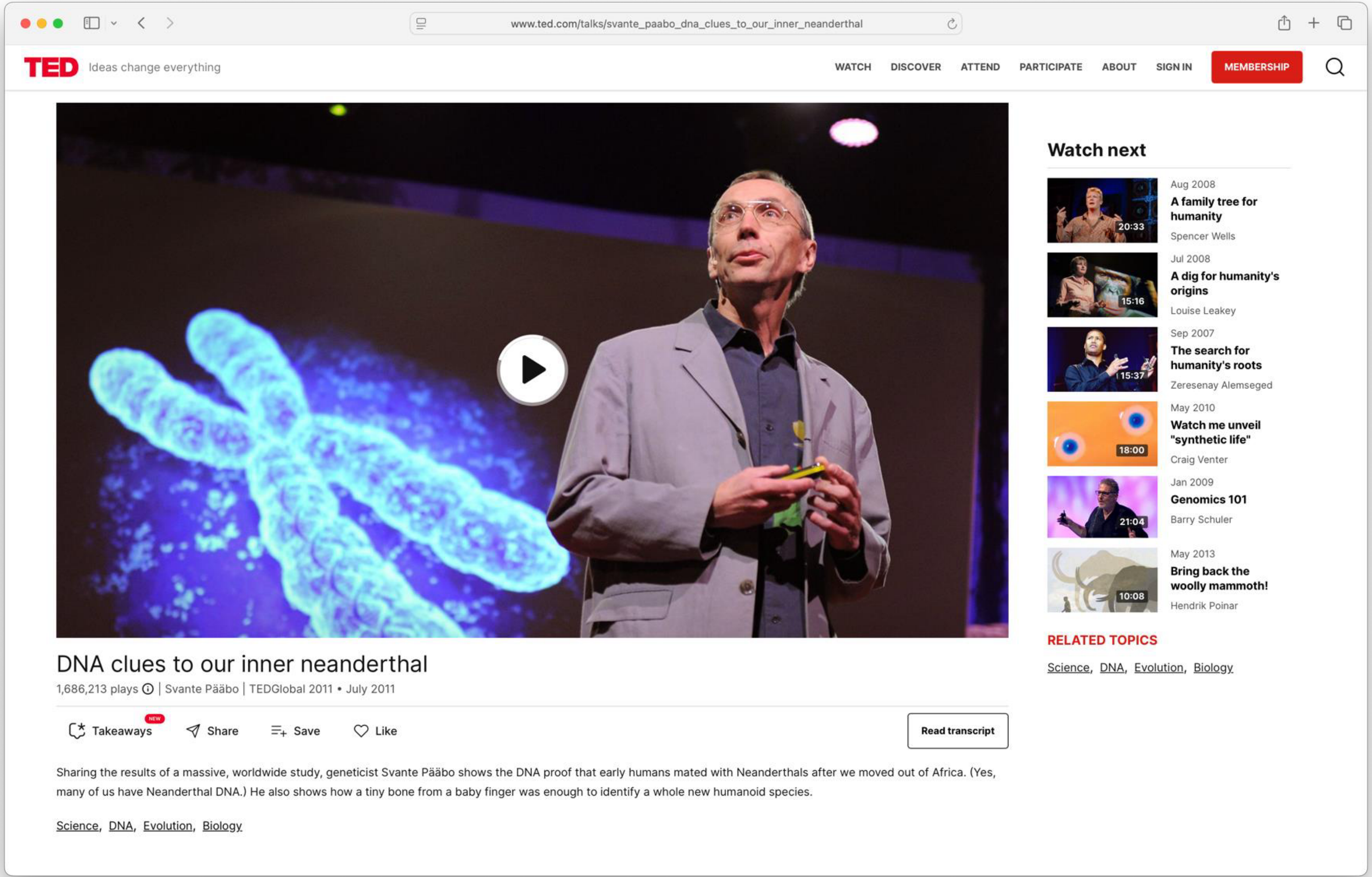


High-coverage whole-genome sequencing of the expanded 1000 Genomes Project cohort including 602 trios
Byrska-Bishop et al. (2022) Cell. doi: 10.1016/j.cell.2022.08.004

Human Migrations

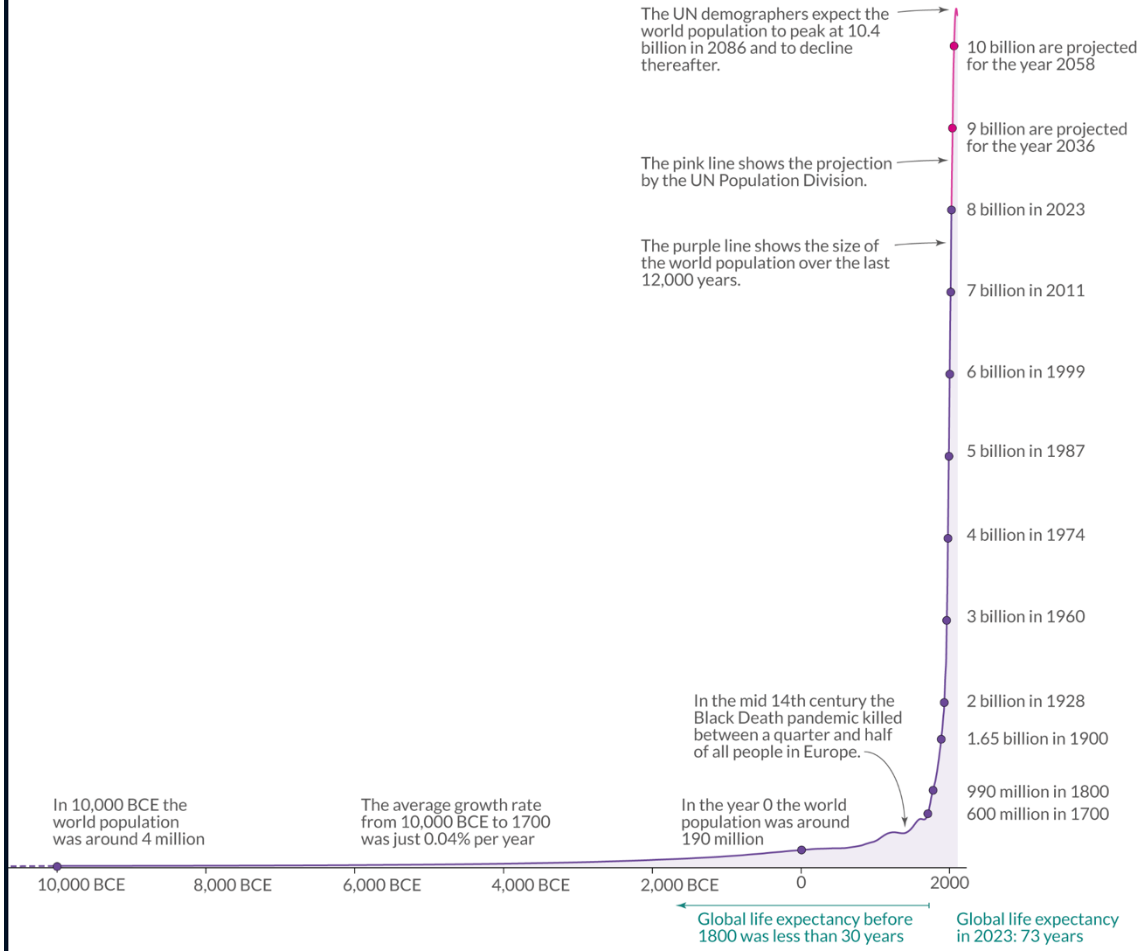


As modern humans migrated out of Africa, they apparently interbred with Neanderthal's so we see their alleles across the rest of the world and carry about 2.5% of their genome with us!



https://www.ted.com/talks/svante_paabo_dna_clues_to_our_inner_neanderthal

The size of the world population over the long-run



Based on estimates by the History Database of the Global Environment (HYDE) and the United Nations.
This is a visualization from OurWorldinData.org.

Our World in Data

The size of the world population over the long-run

Science

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HOME > SCIENCE > VOL. 336, NO. 6082 > RECENT EXPLOSIVE HUMAN POPULATION GROWTH HAS RESULTED IN AN EXCESS OF RARE GENETIC VARIANTS

REPORT

Recent Explosive Human Population Growth Has Resulted in an Excess of Rare Genetic Variants

ALON KEINAN AND ANDREW G. CLARK Authors Info & Affiliations

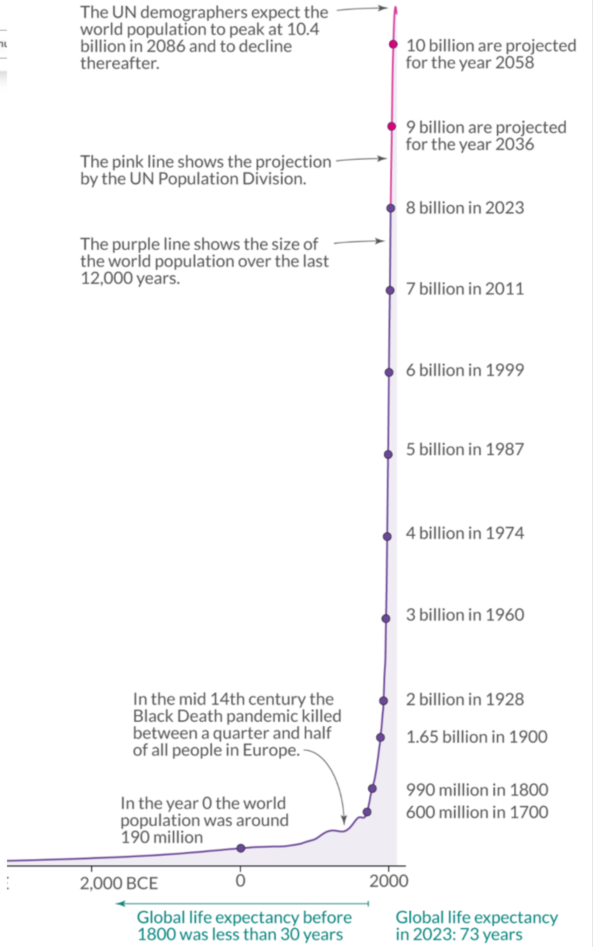
SCIENCE • 11 May 2012 • Vol 336, Issue 6082 • pp. 740-743 • DOI: 10.1126/science.1217283

Exponential Growth Effects

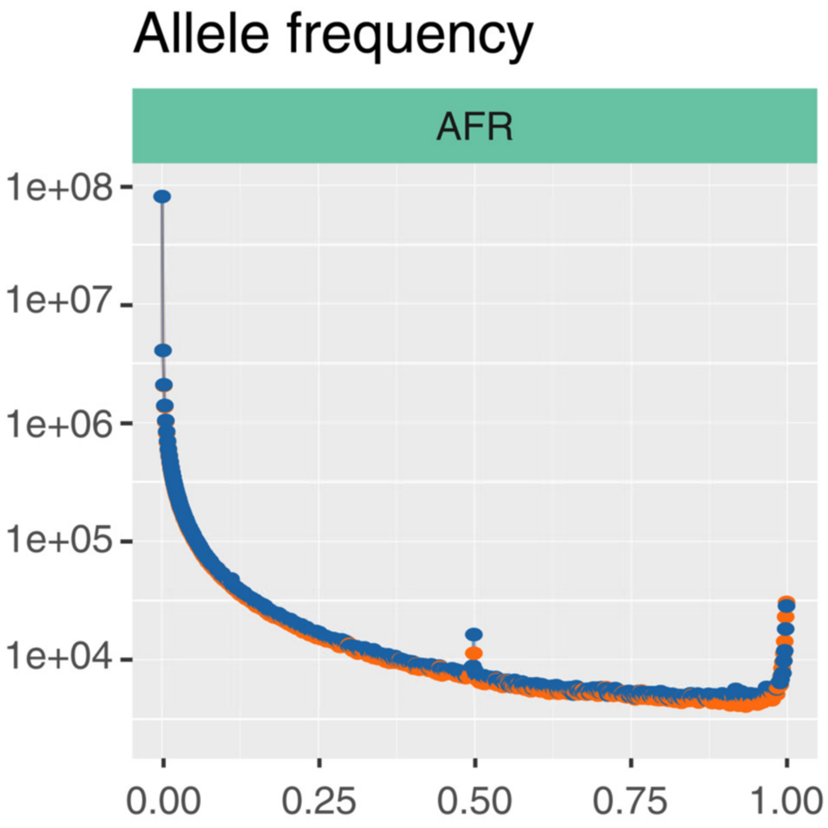
Humans are an extraordinarily successful species, as measured by our large population size—approximately 7 billion—much of which can be put down to recent explosive growth. Leveraging human genomic data, **Keinan and Clark** (p. 740) examined the effects of population growth on our ability to detect rare genetic variants, those hypothesized to be most likely associated with disease. It appears that rapid recent growth increases the load of rare variants and is likely to play an important role in the individual genetic burden of complex disease risk.

Abstract

Human populations have experienced recent explosive growth, expanding by at least three orders of magnitude over the past 400 generations. This departure from equilibrium skews patterns of genetic variation and distorts basic principles of population genetics. We characterized the empirical signatures of explosive growth on the site frequency spectrum and found that the discrepancy in rare variant abundance across demographic modeling studies is mostly due to differences in sample size. Rapid recent growth increases the load of rare variants and is likely to play a role in the individual genetic burden of complex disease risk. Hence, the extreme recent human population growth needs to be taken into consideration in studying the genetics of complex diseases and traits.



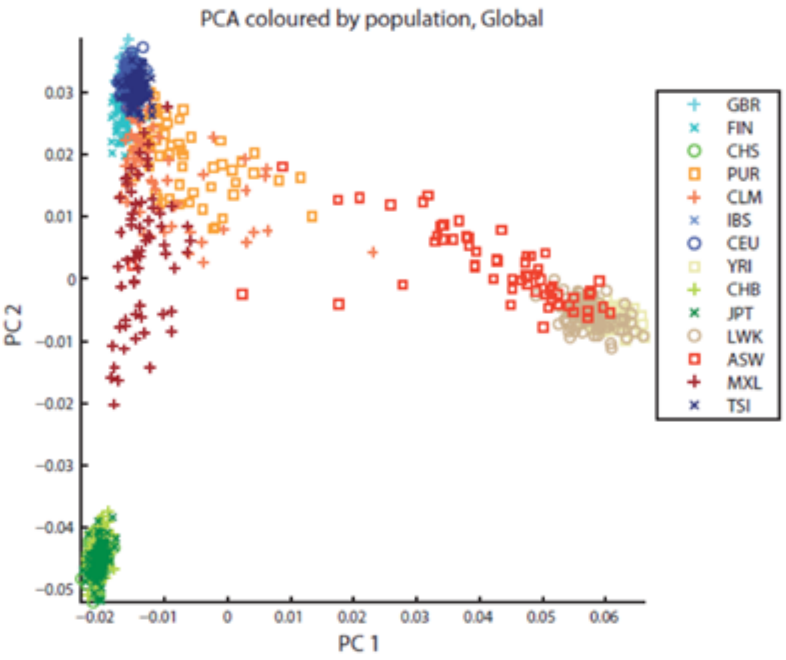
YDE) and the United Nations.



Site frequency spectrum of variants in ~500 AFR genomes from the 1000 Genomes collection

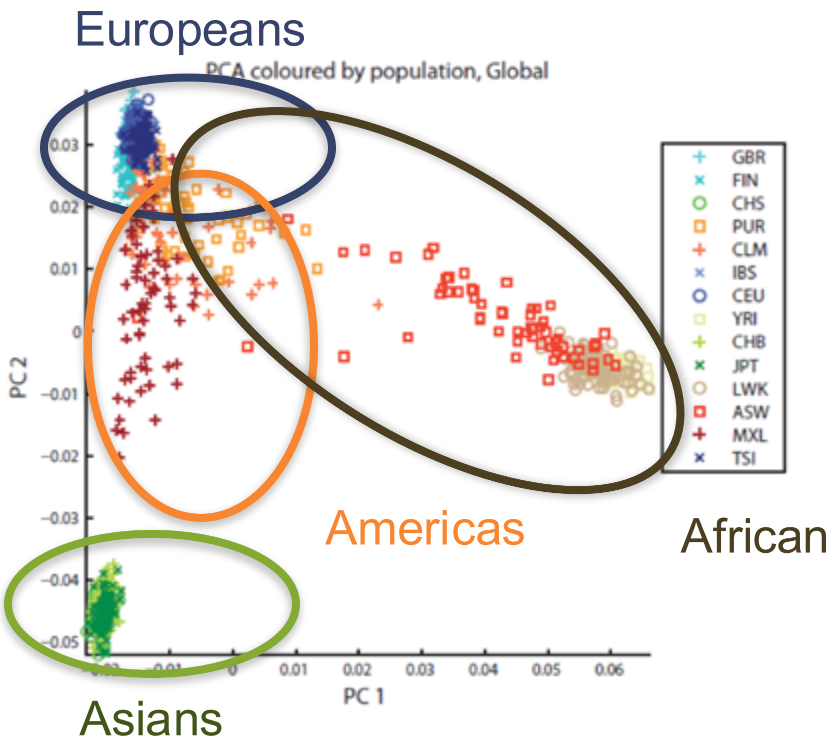
Note the huge fraction of rare alleles (occurring in less than 5% of genomes)

Variation across populations



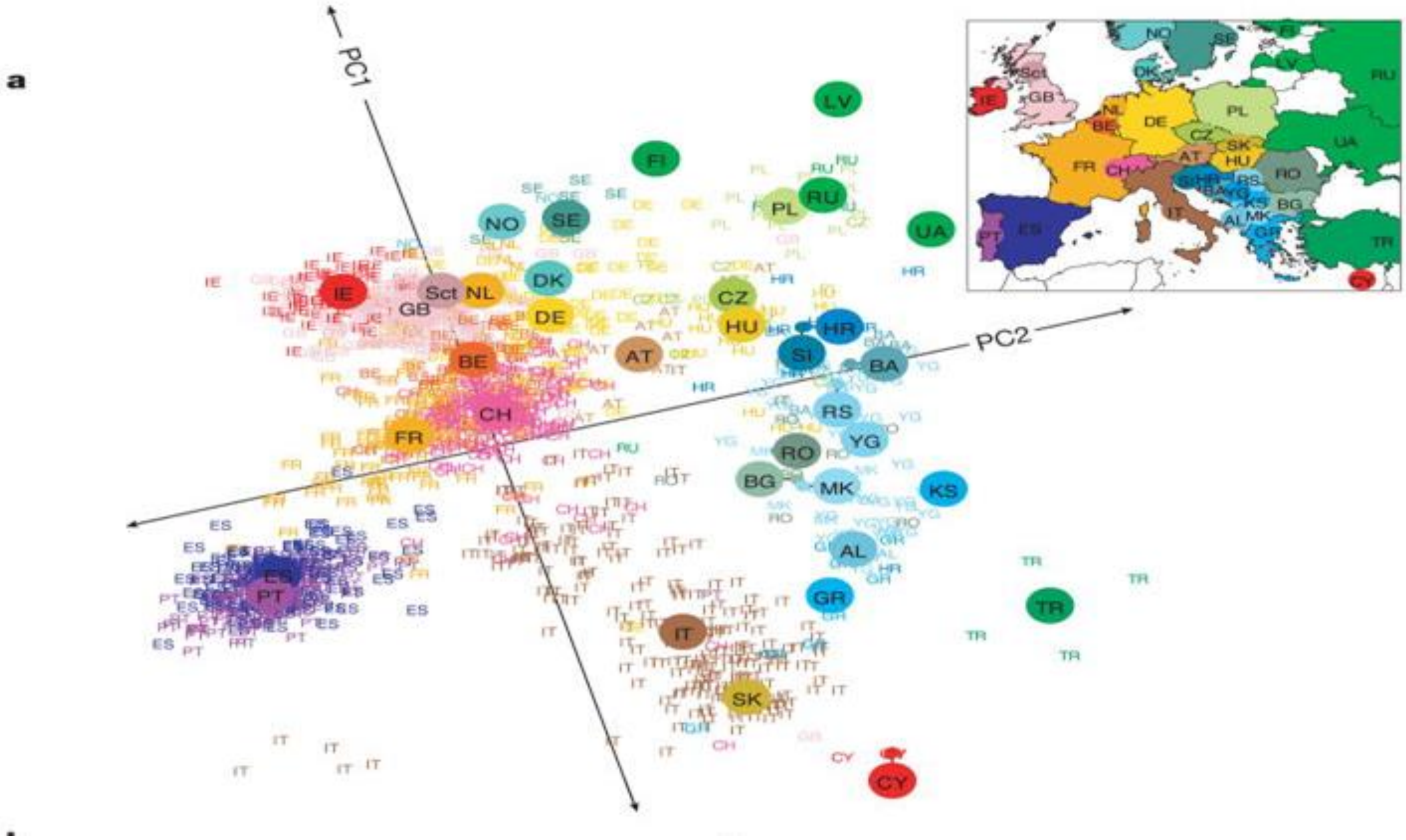
	Person1	Person2	Person3	…
SNP1	0/0	0/1	0/1	
SNP2	0/1	0/1	0/1	
SNP3	0/0	0/0	1/1	
SNP4	1/1	0/1	1/1	
SNP5	0/0	1/1	1/1	
SNP5	0/0	0/0	0/1	
…				

Variation across populations



	Person1	Person2	Person3	…
SNP1	0/0	0/1	0/1	
SNP2	0/1	0/1	0/1	
SNP3	0/0	0/0	1/1	
SNP4	1/1	0/1	1/1	
SNP5	0/0	1/1	1/1	
SNP5	0/0	0/0	0/1	
…				

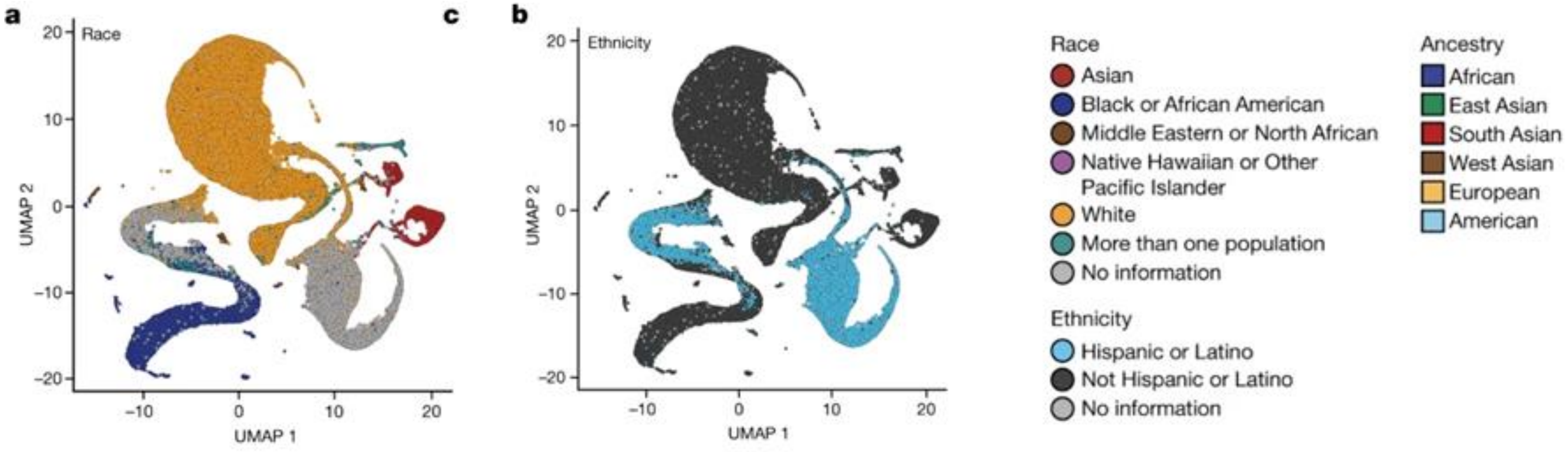
PC1 is largely correlated with African ancestry:
Human genetic diversity is dominated by African ancestry



Genes mirror geography within Europe

Novembre et al (2008) Nature. doi: 10.1038/nature07331

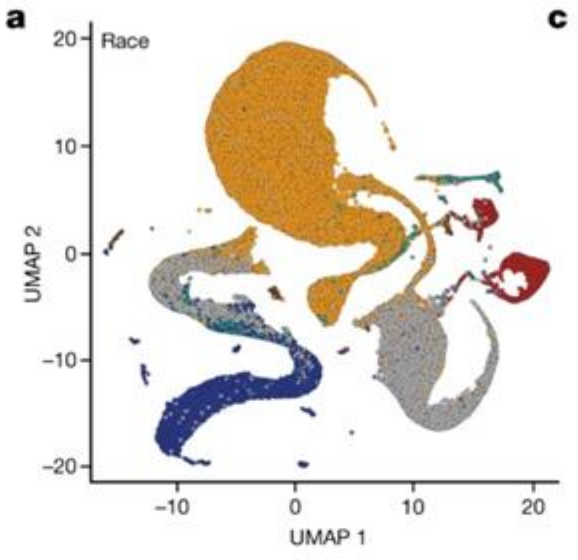
UMAP of Human Variation



Genomic data in the All of Us Research Program

The All of Us Research Program Genomics Investigators (2024) Nature. doi.org/10.1038/s41586-023-06957-x

~~UMAP of Human Variation~~



"It's a pity that All of Us used UMAP to visualize ancestry variation in their new marker paper, out today in Nature. The UMAP algorithm, by design, exaggerates the distinctiveness of the most frequent ancestries, a message that can be misinterpreted by the public. UMAP pulls unusual genotypes towards the majority clusters; in particular it fails to represent admixture in a sensible way (admixture is fundamentally additive, while UMAP is not). In this setting the messiness of Admixture or PCA plots yield a better reflection of the data"

Jonathan Pritchard
Stanford

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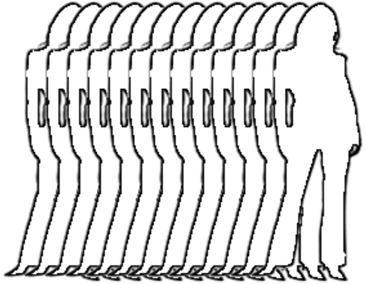
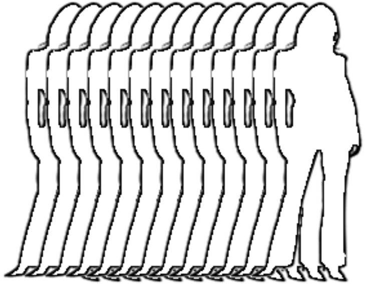
Genome Wide Association (GWAS)





SNP1

Cases
Count of G:
2104 of 4000

Frequency of G:
52.6%

Controls
Count of G:
2676 of 6000

Frequency of G:
44.6%

SNP2

Cases
Count of G:
1648 of 4000

Frequency of G:
41.2%

Controls
Count of G:
2532 of 6000

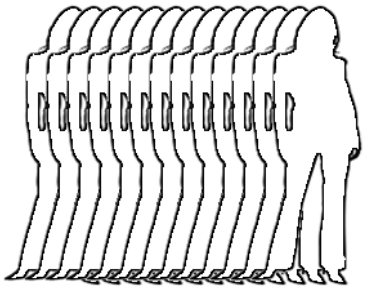
Frequency of G:
42.2%

SNP ...

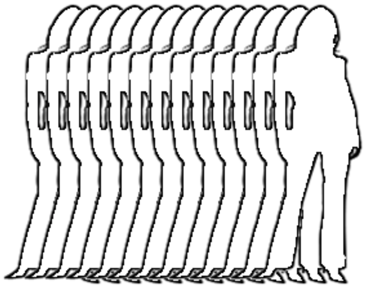
Repeat for all SNPs

Are these significant differences in frequencies?

Genome Wide Association (GWAS)



GC CC GG GC CC GC GC
GG CC GC GG GC GG



GC CC GC GC GG CC CC
CC GC GC GG GC GG

SNP1	*SNP2*	*SNP ...*
Cases	**Cases**	*Repeat for all SNPs*
Count of G: 2104 of 4000	Count of G: 1648 of 4000	
Frequency of G: 52.6%	Frequency of G: 41.2%	
Controls	**Controls**	
Count of G: 2676 of 6000	Count of G: 2532 of 6000	
Frequency of G: 44.6%	Frequency of G: 42.2%	
P-value: $5.0 \cdot 10^{-15}$	**P-value:** 0.33	

Pearson’s Chi-squared test

The value of the test-statistic is

$$\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i} = N \sum_{i=1}^{n} \frac{(O_i/N - p_i)^2}{p_i}$$

where

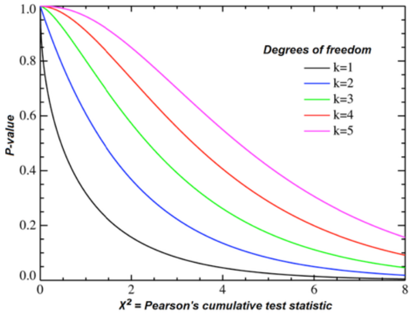
χ^2 = Pearson's cumulative test statistic, which asymptotically approaches a χ^2 distribution.

O_i = the number of observations of type *i*.

N = total number of observations

$E_i = Np_i$ = the expected (theoretical) frequency of type *i*, asserted by the null hypothesis that the fraction of type *i* in the population is p_i

n = the number of cells in the table.



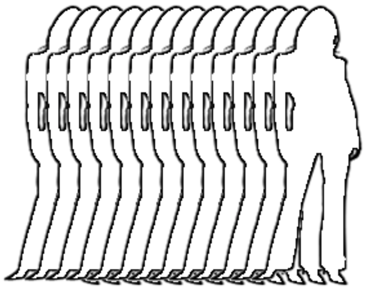
$$P(\chi_P^2(\{p_i\}) > T) \sim C \int_{\sum_{i=1}^{m-1} y_i^2 > T} \left\{ \prod_{i=1}^{m-1} dy_i \right\} \prod_{i=1}^{m-1} \exp\left[-\frac{1}{2} \left(\sum_{i=1}^{m-1} y_i^2 \right) \right]$$

	has G	Not G	*Marginal Row Totals*
Cases	2104 (1912) [19.28]	1896 (2088) [17.66]	4000
Controls	2676 (2868) [12.85]	3324 (3132) [11.77]	6000
Marginal Column Totals	4780	5220	10000 (Grand Total)

Cases/hasG expected: 4000 * (4780/10000) = 1912 expected
Cases/hasG squared deviation: $(2104 – 1912)^2$ / 1912 = 19.28 deviation

The chi-square statistic is 19.28+17.66+12.85+11.77 = 61.56. The p-value is 5e-15

Genome Wide Association (GWAS)



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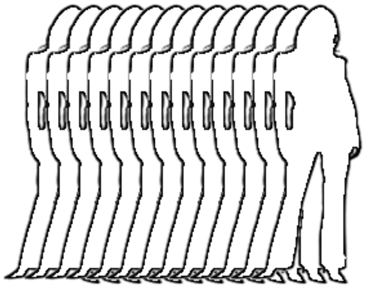

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SNP ...

Repeat for all SNPs

Chi-squared or similar test

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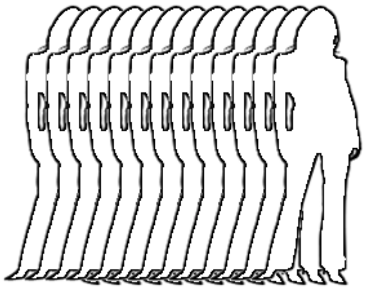

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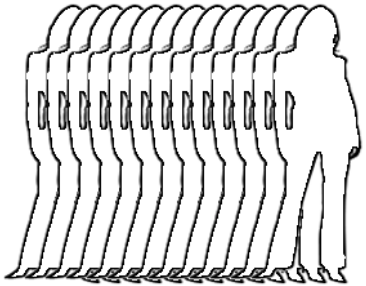

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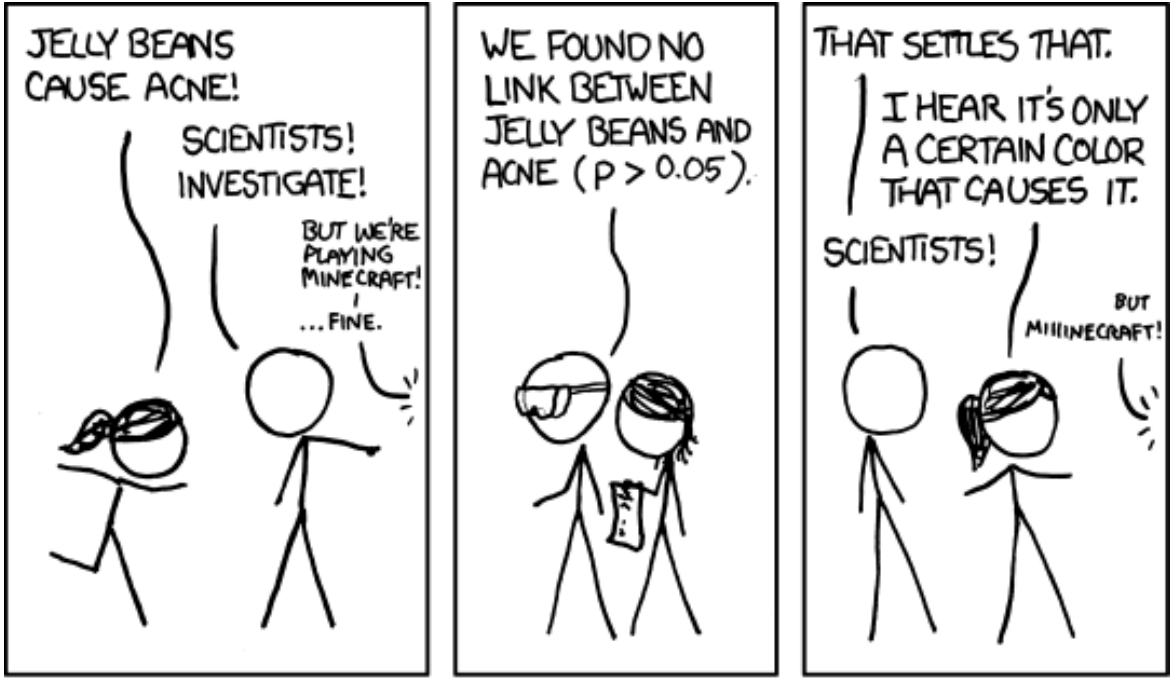
With a (much) larger population, this might be a significant difference in rate:
25320/60000 =>
p = 5e-7

Chi-squared or similar test

The curse of multiple testing



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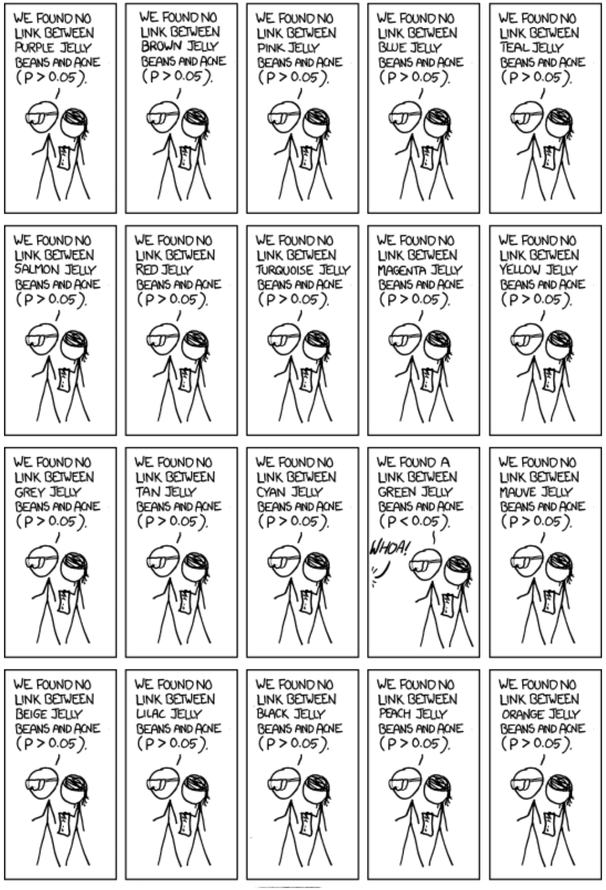

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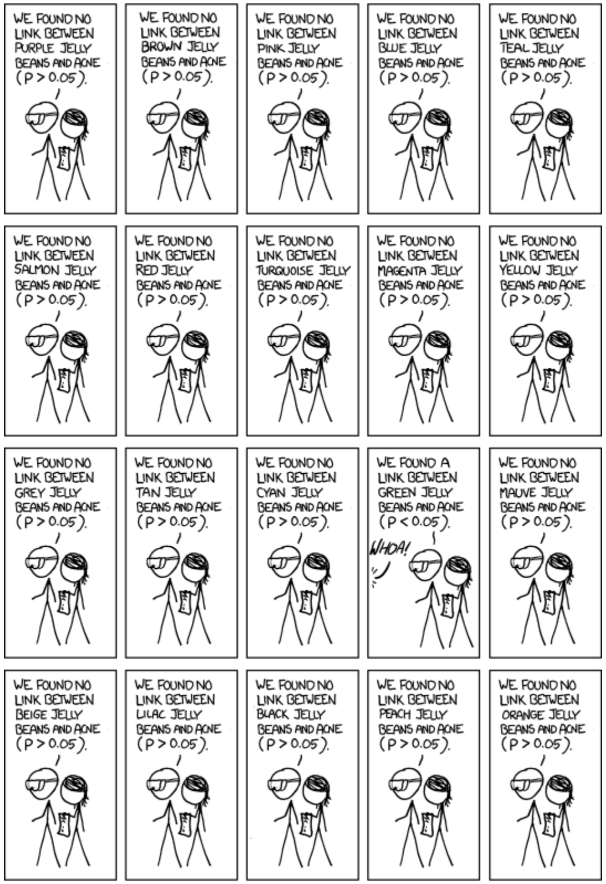
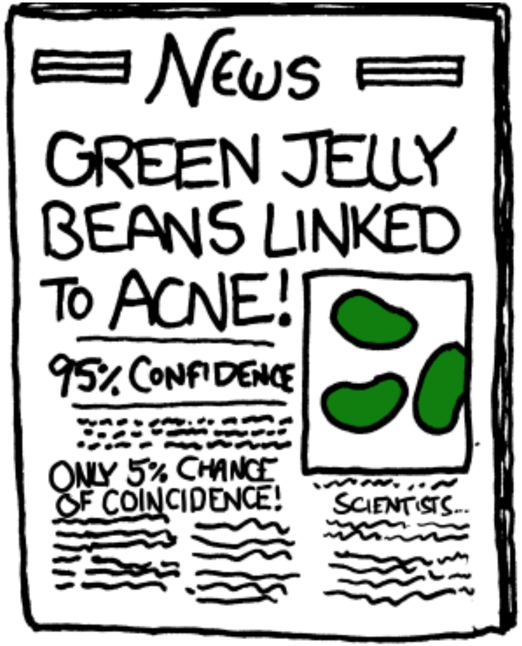
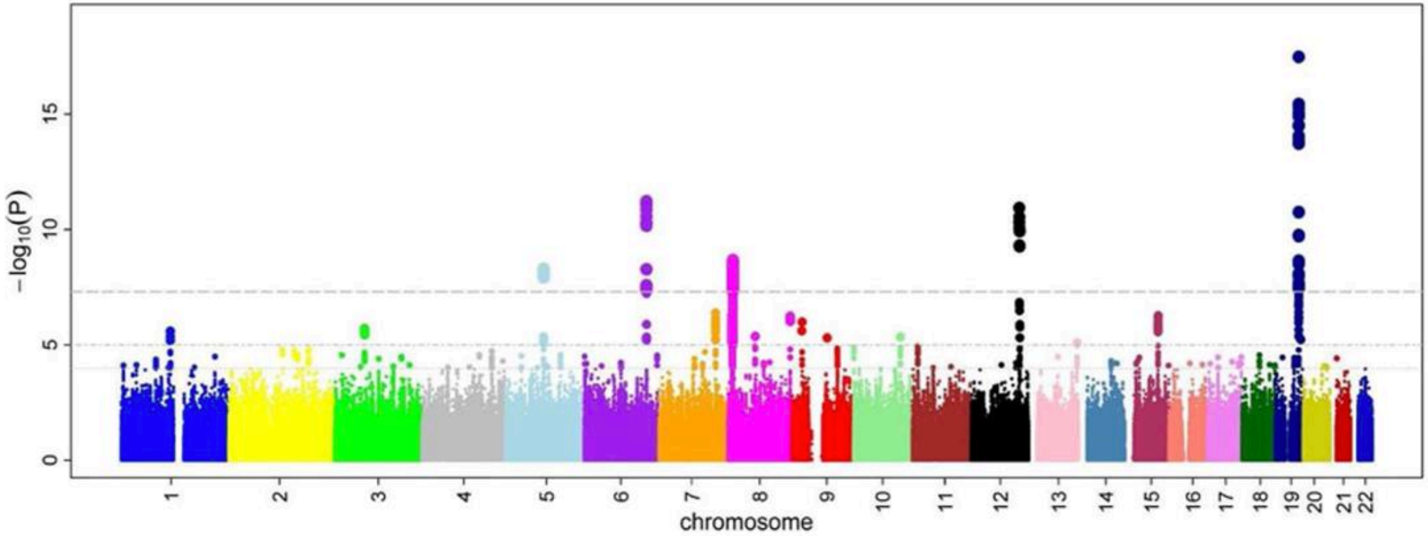




Manhattan Plot



Four Novel Loci (19q13, 6q24, 12q24, and 5q14) Influence the Microcirculation In Vivo

Ikram et al (2010) PLOS Genetics. doi: 10.1371/journal.pgen.1001184

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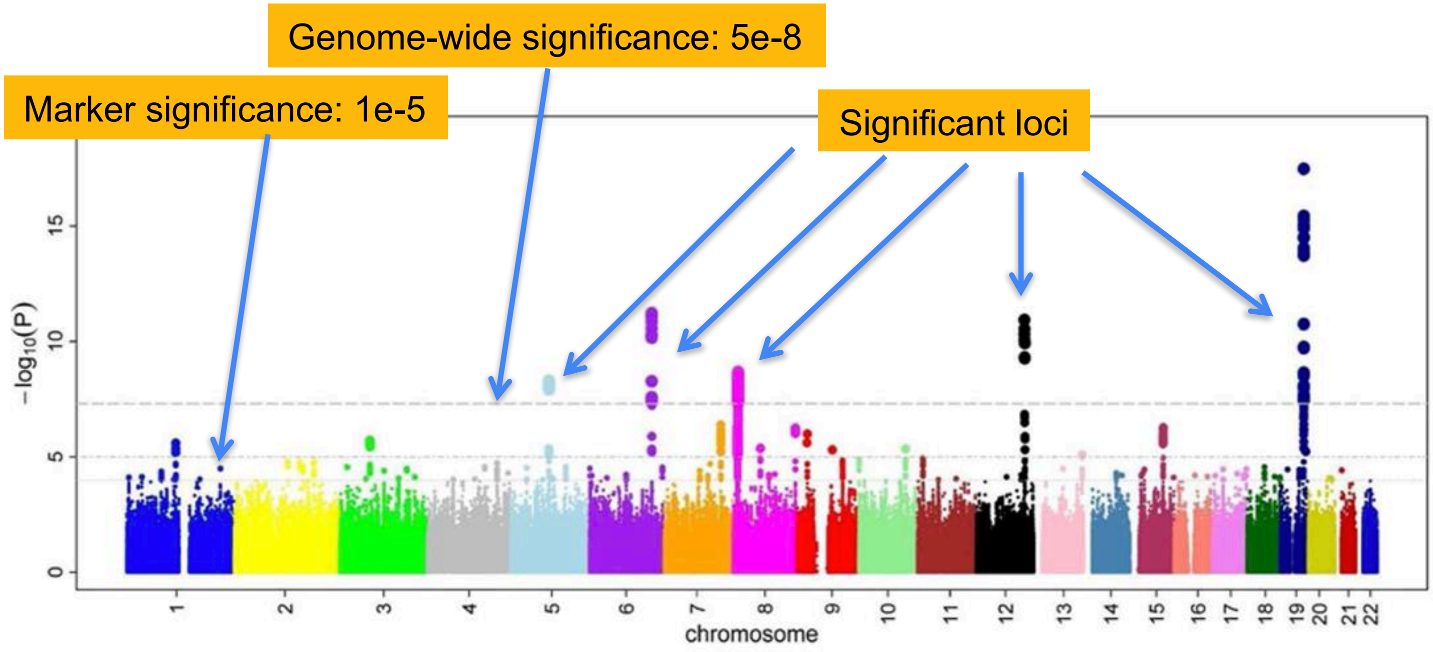

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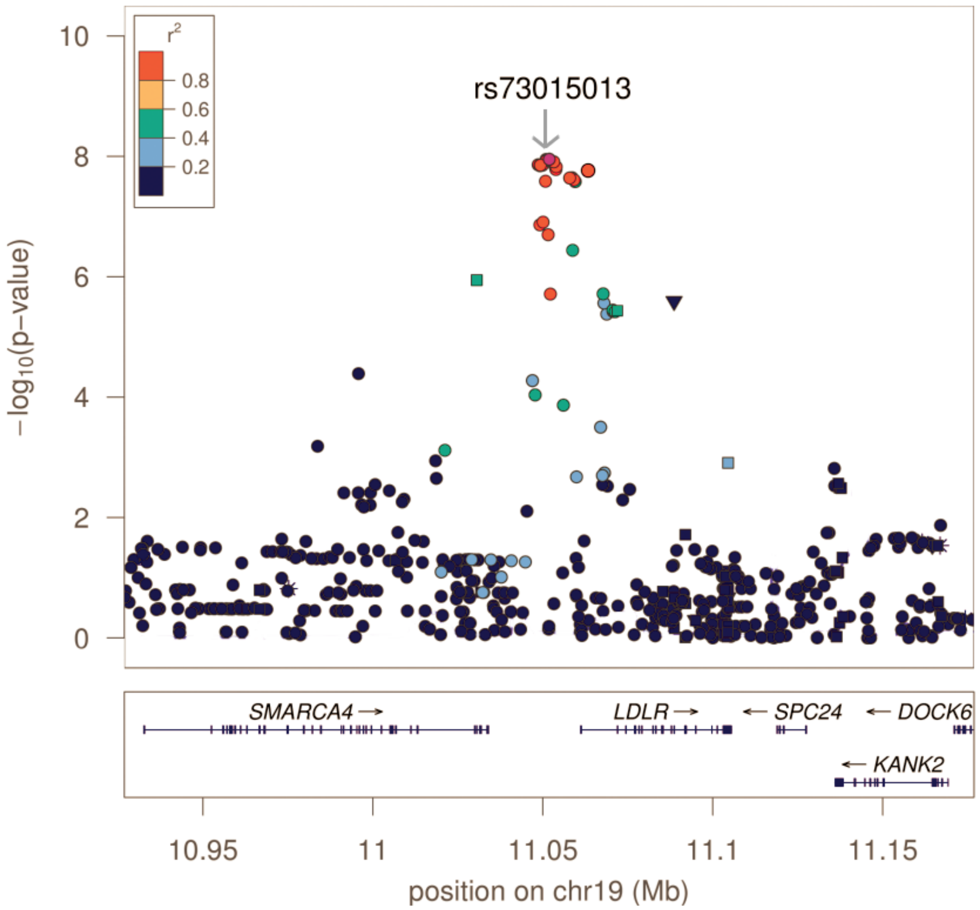
Regional Association Plot



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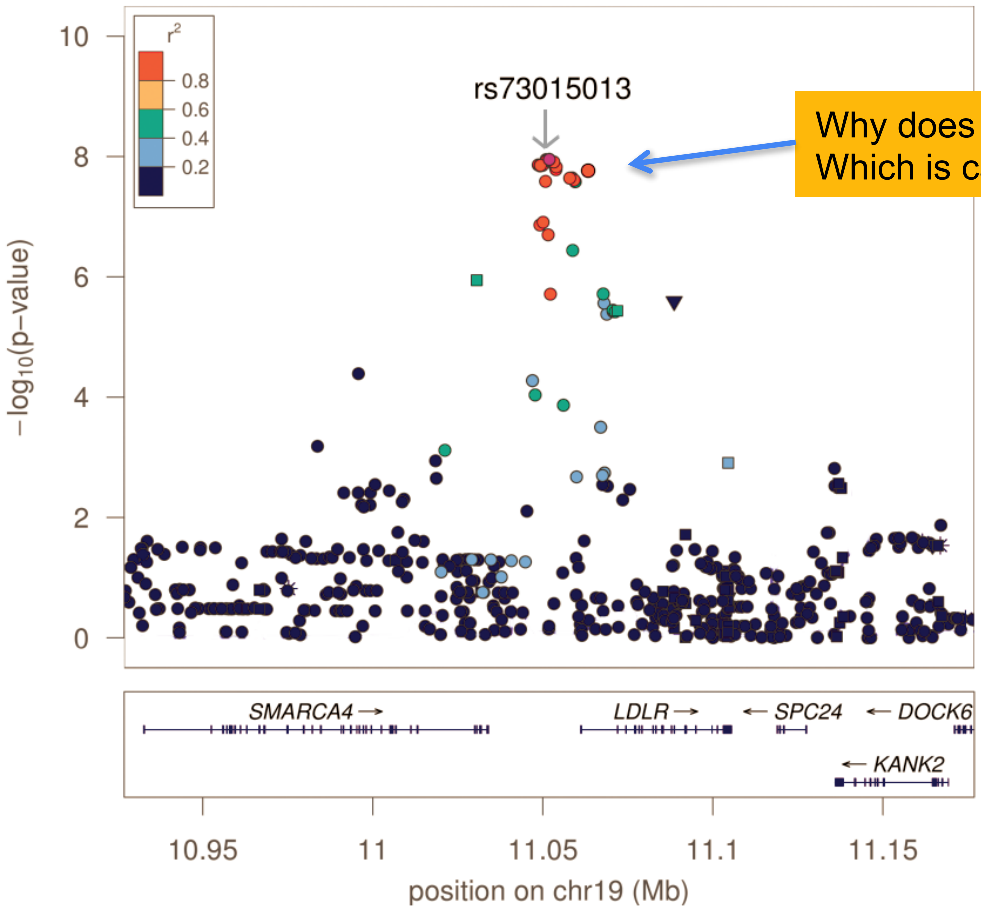

Why does it look like this?
Which is causal?

First published GWAS

Complement Factor H Polymorphism in Age-Related Macular Degeneration

Robert J. Klein,[1] Caroline Zeiss,[2*] Emily Y. Chew,[3*] Jen-Yue Tsai,[4*] Richard S. Sackler,[1] Chad Haynes,[1] Alice K. Henning,[5] John Paul SanGiovanni,[3] Shrikant M. Mane,[6] Susan T. Mayne,[7] Michael B. Bracken,[7] Frederick L. Ferris,[3] Jurg Ott,[1] Colin Barnstable,[2] Josephine Hoh[7]†

Age-related macular degeneration (AMD) is a major cause of blindness in the elderly. We report a genome-wide screen of 96 cases and 50 controls for polymorphisms associated with AMD. Among 116,204 single-nucleotide polymorphisms genotyped, an intronic and common variant in the complement factor H gene (*CFH*) is strongly associated with AMD (nominal *P* value $<10^{-7}$). In individuals homozygous for the risk allele, the likelihood of AMD is increased by a factor of 7.4 (95% confidence interval 2.9 to 19). Resequencing revealed a polymorphism in linkage disequilibrium with the risk allele representing a tyrosine-histidine change at amino acid 402. This polymorphism is in a region of CFH that binds heparin and C-reactive protein. The *CFH* gene is located on chromosome 1 in a region repeatedly linked to AMD in family-based studies.

Age-related macular degeneration (AMD) is the leading cause of blindness in the developed world. Its incidence is increasing as the elderly population expands (*1*). AMD is characterized by progressive destruction of the retina's central region (macula), causing central field visual loss (*2*). A key feature of AMD is the formation of extracellular deposits called drusen concentrated in and around the macula behind the retina between the retinal pigment epithelium (RPE) and the choroid. To date, no therapy for this disease has proven to be broadly effective. Several risk factors have been linked to AMD, including age, smoking, and family history (*3*). Candidate-gene studies have not found any genetic differences that can account for a large proportion of the overall prevalence (*2*). Family-based whole-genome linkage scans have identified chromosomal regions that show evidence of linkage to AMD (*4–8*), but the linkage areas have not been resolved to any causative mutations.

Like many other chronic diseases, AMD is caused by a combination of genetic and environmental risk factors. Linkage studies are not as powerful as association studies for the identification of genes contributing to the risk for common, complex diseases (*9*). However, linkage studies have the advantage of searching the whole genome in an unbiased manner without presupposing the involvement of particular genes. Searching the whole genome in an association study requires typing 100,000 or more single-nucleotide polymorphisms (SNPs) (*10*). Because of these technical demands, only one whole-genome association study, on susceptibility to myocardial infarction, has been published to date (*11*).

Study design. We report a whole-genome case-control association study for genes involved in AMD. To maximize the chance of success, we chose clearly defined phenotypes for cases and controls. Case individuals exhibited at least some large drusen in a quantitative photographic assessment combined with evidence of sight-threatening AMD (geographic atrophy or neovascular AMD). Control individuals had either no or only a few small drusen. We analyzed our data using a statistically conservative approach to correct for the large number of SNPs tested, thereby guaranteeing that the probability of a false positive is no greater than our reported *P* values.

We used a subset of individuals who participated in the Age-Related Eye Disease Study (AREDS) (*12*). From the AREDS

[1]Laboratory of Statistical Genetics, Rockefeller University, 1230 York Avenue, New York, NY 10021, USA. [2]Department of Ophthalmology and Visual Science, Yale University School of Medicine, 330 Cedar Street, New Haven, CT 06520, USA. [3]National Eye Institute, Building 10, CRC, 10 Center Drive, Bethesda, MD 20892–1204, USA. [4]Biological Imaging Core, National Eye Institute, 9000 Rockville Pike, Bethesda, MD 20892, USA. [5]The EMMES Corporation, 401 North Washington Street, Suite 700, Rockville MD 20850, USA. [6]W. M. Keck Facility, Yale University, 300 George Street, Suite 201, New Haven, CT 06511, USA. [7]Department of Epidemiology and Public Health, Yale University School of Medicine, 60 College Street, New Haven, CT 06520, USA.

*These authors contributed equally to this work.
†To whom correspondence should be addressed. E-mail: josephine.hoh@yale.edu

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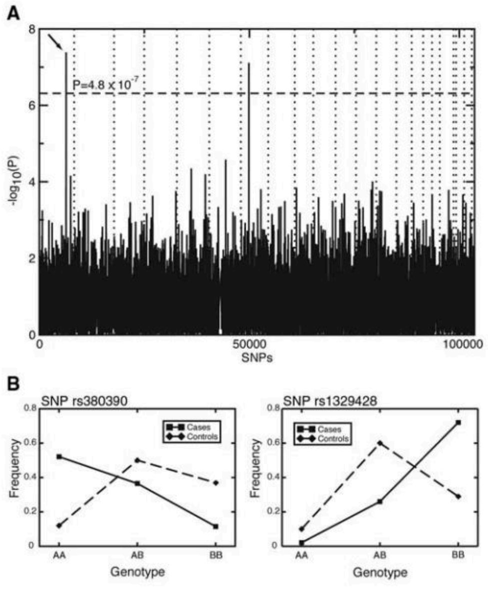

Fig. 1. (**A**) *P* values of genome-wide association scan for genes that affect the risk of developing AMD. $-\log_{10}(p)$ is plotted for each SNP in chromosomal order. The spacing between SNPs on the plot is uniform and does not reflect distances between SNPs on the chromosomes. The dotted horizontal line shows the cutoff for $P = 0.05$ after Bonferroni correction. The vertical dotted lines show chromosomal boundaries. The arrow indicates the peak for SNP rs380390, the most significant association, which was studied further. (**B**) Variation in genotype frequencies between cases and controls.

GWAS Catalog

As of 2025-09-15, the GWAS Catalog contains 7410 publications, 1,008,330 top associations and 139,254 full summary statistics.

http://www.ebi.ac.uk/gwas/diagram